# 2. Prediction Spaces, Calibration and Sharpness

# Disclaimer

We will now turn to a formal, measure-theoretic introduction of prediction spaces, which are probability spaces tailored to the study of forecasting.

We will then discuss the critical notions of calibration, dispersion, and sharpness, and argue that the goal in probabilistic forecasting is to maximize the sharpness of the predictive distributions, subject to calibration.

The exposition largely follows Gneiting and Ranjan (2013).

In applications, the measure-theoretic construction will remain hypothetical.

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# Prediction spaces

Consider the joint distribution of multiple probabilistic forecasts and the observation on a probability space

$$(\Omega, \mathcal{A}, \mathbb{Q}).$$

We assume that the elements of the sample space $\Omega$ can be identified with tuples of the form

$$(P_1, \ldots, P_k, Y),$$

where each of $P_1, \ldots, P_k$ is a probability measure on the outcome space for the observation, $Y$,

$$(\Omega_Y, \mathcal{A}_Y).$$

The probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{A})$ specifies the joint distribution of the probabilistic forecasts and the observation.

# Information sets

For $i = 1, \ldots, k$, we require the random probability measure $P_i$ to be measurable with respect to the sub-$\sigma$-algebra $\mathcal{A}_i \subseteq \mathcal{A}$ that encodes the forecast's information set:

- consists of data, expertise, theories and assumptions at hand

The probabilistic forecasts $P_1, \ldots, P_k$ might stem from distinct experts, organizations or statistical models, as commonly encountered in the practice of forecasting.

# Notation

In what follows in the present section,

- we restrict the discussion to the case of a real-valued observation, so that $(\Omega_Y, \mathcal{A}_Y) = (\mathbb{R}, \mathcal{B})$
- a probabilistic forecast then corresponds to (a Lebesgue-Stieltjes probability measure on the real line, which we identify with the associated) right-continuous CDF
- We write $\sigma(\mathcal{A}_1, \ldots, \mathcal{A}_m)$ and $\sigma(X_1, \ldots, X_n)$ to denote the $\sigma$-algebra generated by the families $\mathcal{A}_1, \ldots, \mathcal{A}_m$ of subsets of $\Omega$, or by the random variables $X_1, \ldots, X_n$, respectively.
- We use the symbol $\mathcal{L}$ to denote an unconditional or conditional distribution.

# Prediction space

**Definition 2.1: Prediction space**

Let $k \geq 1$ be an integer. A **prediction space** is a probability space $(\Omega, \mathcal{A}, \mathbb{Q})$ together with sub-$\sigma$-algebras $\mathcal{A}_1, \ldots, \mathcal{A}_k \subseteq \mathcal{A}$, where the elements of the sample space $\Omega$ can be identified with tuples

$$(F_1, \ldots, F_k, Y, V)$$

such that

(P1) for $i = 1, \ldots, k$, $F_i$ is a CDF-valued random quantity that is measurable with respect to the sub-$\sigma$-algebra $\mathcal{A}_i$,

(P2) $Y$ is a real-valued random variable,

(P3) $V$ is a random variable that is uniformly distributed on the unit interval and independent of $\mathcal{A}_1, \ldots, \mathcal{A}_k$ and $Y$.

# Ideal probabilistic forecasts

Subsequent definitions and phrases such as *almost surely* or *with positive probability* refer to the probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{A})$ that determines the joint distribution of the probabilistic forecasts and the observations.

The sub-$\sigma$-algebra $\mathcal{A}_i$ encodes the information set for the CDF-valued random quantity $F_i$ which may, but need not, be ideal in the following sense.

**Definition 2.2**

The CDF-valued random quantity $F_i$ is **ideal** relative to the sub-$\sigma$-algebra $\mathcal{A}_i$ if

$$F_i = \mathcal{L}(Y \mid \mathcal{A}_i)$$

almost surely.

# Example: Binary events

We consider probability forecasts of a dichotomous outcome, such as rain *vs.* no rain, or recession *vs.* no recession.

We identify a success or occurrence with the outcome $Y = 1$, and a non-success with $Y = 0$.

The CDF-valued random quantity $F_i$, where $i = 1, \ldots, k$, thus is of the form

$$F_i(y) = (1 - p_i)\mathbf{1}\{y \geq 0\} + p_i\mathbf{1}\{y \geq 1\}$$

and can be identified with the probability forecast $p_i$ for a success, where

$$p_i \in [0, 1]$$

is a random variable.

# Example: Density forecasts

Typically, it suffices to consider the joint distribution of the tuple $(F_1, \ldots, F_k, Y)$, without any need to explicitly specify other facets of the prediction space.

To define a prediction space, let

$$Y \mid \mu \sim \mathcal{N}(\mu, 1) \qquad \text{where} \qquad \mu \sim \mathcal{N}(0, 1),$$

and let $\tau$ attain the values 1 and $-1$ with equal probability, independently of $\mu$ and $Y$, i.e.,

$$\mathbb{P}(\tau = 1) = \mathbb{P}(\tau = -1) = \frac{1}{2}.$$

# Quiz

In this example, a forecaster has access to the value of $\mu$. Which predictive distribution is ideal relative to the $\sigma$-algebra $\sigma(\mu)$ generated by $\mu$?

A: $\mathcal{N}(\mu, 1)$

B: $\mathcal{N}(0, 2)$

C: $\mathcal{N}(\mu, 2)$

D: $\mathcal{N}(0, 1)$

# Example: Density forecasts

Consider the following four density forecasts:

| Forecast | Predictive Distribution |
|---|---|
| Perfect | $F_1 = \mathcal{N}(\mu, 1)$ |
| Climatological | $F_2 = \mathcal{N}(0, 2)$ |
| Unfocused | $F_3 = \frac{1}{2}\left(\mathcal{N}(\mu, 1) + \mathcal{N}(\mu + \tau, 1)\right)$ |
| Sign-reversed | $F_4 = \mathcal{N}(-\mu, 1)$ |

- The perfect forecast is ideal relative to the sub-$\sigma$-algebra generated by $\mu$.
- The climatological forecast is ideal relative to the trivial sub-$\sigma$-algebra.
- The unfocused and sign-reversed forecasts are not ideal.

Details and further properties: Exercises

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# Calibration: Motivation

Intuitively, a probabilistic forecast is calibrated if it is statistically consistent with the observations:

*The observation should look like a random draw from the predictive distribution.*

Example: If a weather service issues a 30% chance of rain on 100 days, we expect rain on about 30 of these days.

Calibration is a property of the joint distribution of forecasts and observations, and can only be assessed from a sample of forecast cases.

In the following, we introduce several notions of calibration and dispersion, and diagnostic tools for checking them (Gneiting et al., 2007; Gneiting and Ranjan, 2013).

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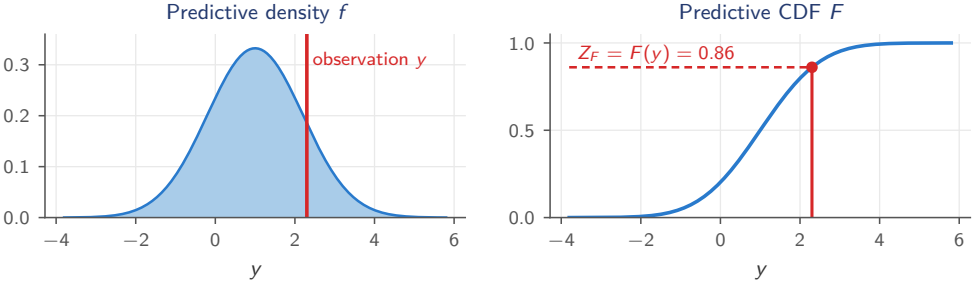
2.3 Sharpness

# Probability integral transform

If $F$ denotes a fixed, non-random predictive CDF for an observation $Y$, the probability integral transform (PIT) is the random variable

$$Z_F = F(Y),$$

i.e., the value that the predictive CDF attains at the observation.



# Probability integral transform

It is well known that if $F$ is continuous and $Y \sim F$, then $Z_F$ is standard uniform.

Proof (for strictly increasing $F$): For $z \in (0, 1)$,

$$\mathbb{Q}(Z_F \leq z) = \mathbb{Q}(F(Y) \leq z) = \mathbb{Q}(Y \leq F^{-1}(z)) = F(F^{-1}(z)) = z.$$

However: If $F$ has jumps, then $F(Y)$ is not uniform, even if $Y \sim F$.

Example: If $Y \sim \text{Ber}(p)$, then $F(Y)$ only attains the values $1 - p$ and 1.

Predictive distributions with jumps are common in practice, e.g., for

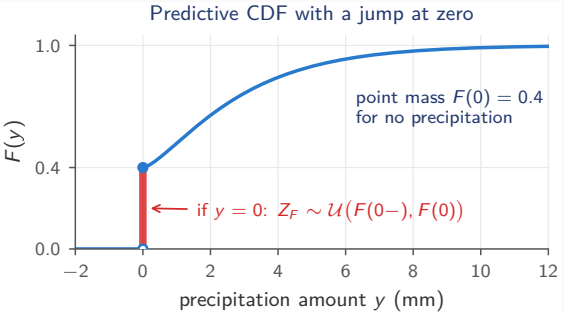
- count data (e.g., the number of goals in a soccer match),
- precipitation amounts (point mass at zero),
- ensemble forecasts, which are identified with their empirical CDF.

# Randomized probability integral transform

Let $V$ be standard uniform and independent of $Y$, and write $F(y-) = \lim_{x \uparrow y} F(x)$ for the left-hand limit of $F$ at $y \in \mathbb{R}$. The randomized PIT

$$Z_F = F(Y-) + V\,(F(Y) - F(Y-))$$

draws a uniform value from the jump of $F$ at the observation (if there is one).




If $Y \sim F$, then the randomized PIT is standard uniform for any fixed CDF $F$ (Rüschendorf, 2009).

Proof: Exercise

# PIT in the prediction space setting

In the prediction space setting, $F$ is a CDF-valued random quantity, and we use the auxiliary variable $V$ from assumption (P3) in Definition 2.1.

**Definition 2.3: Probability integral transform**

In the prediction space setting, the random variable

$$Z_F = F(Y-) + V\,(F(Y) - F(Y-))$$

is the **probability integral transform** (PIT) of the CDF-valued random quantity $F$.

In a nutshell, the PIT is the value that the predictive CDF attains at the observation, with suitable adaptions at any points of discontinuity.

The PIT takes values in the unit interval, so that $\mathrm{Var}(Z_F) \in [0, \frac{1}{4}]$.

# Quiz

The predictive distribution is $F = \mathcal{N}(0, 1)$, and we observe $y = 1.645$. What is the value of the PIT?

A: 0.05

B: 0.5

C: 0.95

D: 1.645

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# Notions of calibration

In what follows, we use the terms CDF-valued random quantity and forecast interchangeably.

**Definition 2.4: Calibration**

In the prediction space setting, let $F$ be a forecast with PIT $Z_F$.

(a) $F$ is **marginally calibrated** if $\mathbb{E}_{\mathbb{Q}}[F(y)] = \mathbb{Q}(Y \leq y)$ for all $y \in \mathbb{R}$.

(b) $F$ is **probabilistically calibrated** if its PIT $Z_F$ is uniformly distributed on the unit interval.

- In the definition of marginal calibration, the left-hand side depends on the law of the forecast, whereas the right-hand side depends on the law of the observation ("forecast climate = actual climate").
- Probabilistic calibration involves the joint law of the forecast and the observation.

**Definition 2.5: Dispersion**

In the prediction space setting, let $F$ and $G$ be forecasts with PITs $Z_F$ and $Z_G$.

(a) $F$ is **overdispersed** if $\mathrm{Var}(Z_F) < \frac{1}{12}$, **neutrally dispersed** if $\mathrm{Var}(Z_F) = \frac{1}{12}$, and **underdispersed** if $\mathrm{Var}(Z_F) > \frac{1}{12}$.

(b) $F$ is **at least as dispersed** as $G$ if $\mathrm{Var}(Z_F) \leq \mathrm{Var}(Z_G)$, and **more dispersed** than $G$ if $\mathrm{Var}(Z_F) < \mathrm{Var}(Z_G)$.

(c) $F$ is **regular** if the support of the distribution of $Z_F$ is the unit interval.

Intuition: Recall that $\frac{1}{12}$ is the variance of the uniform distribution on $[0, 1]$. If $F$ is too narrow (underdispersed), the observation often falls into its tails, so that PIT values close to 0 or 1 are frequent and $\mathrm{Var}(Z_F)$ is large.

# Calibration and dispersion

Dispersion is defined via the PIT and thus involves the joint law of the forecast and the observation. In contrast, the spread of the predictive distribution itself is a measure of sharpness that does not consider the observation (Section 2.3).

**Lemma 2.6**

A probabilistically calibrated forecast is neutrally dispersed and regular.

The converse is not necessarily true, in that a forecast which is neutrally dispersed need not be calibrated nor regular.

However, an ideal forecast is always calibrated.

# Quiz

For a large sample of forecast cases, the PIT values of a forecasting method have an empirical variance of 0.12. The forecasts are

A: overdispersed

B: neutrally dispersed

C: underdispersed

D: probabilistically calibrated

# Ideal forecasts are calibrated

**Theorem 2.7**

A forecast that is ideal relative to a $\sigma$-algebra is both marginally calibrated and probabilistically calibrated.

Proof: Let $F = \mathcal{L}(Y \mid \mathcal{A}_0)$ be ideal relative to $\mathcal{A}_0$, i.e., $F(y) = \mathbb{Q}(Y \leq y \mid \mathcal{A}_0)$ almost surely for all $y \in \mathbb{R}$.

*Marginal calibration*: By the tower property of conditional expectations,

$$\mathbb{E}_{\mathbb{Q}}[F(y)] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{Q}(Y \leq y \mid \mathcal{A}_0)\right] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{E}_{\mathbb{Q}}\left[\mathbf{1}\{Y \leq y\} \mid \mathcal{A}_0\right]\right] = \mathbb{Q}(Y \leq y).$$

*Probabilistic calibration*: Given $\mathcal{A}_0$, the forecast $F$ is fixed and $Y \sim F$. By the uniformity of the randomized PIT for fixed CDFs, $\mathbb{Q}(Z_F \leq z \mid \mathcal{A}_0) = z$ for $z \in (0, 1)$, and so

$$\mathbb{Q}(Z_F \leq z) = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{Q}(Z_F \leq z \mid \mathcal{A}_0)\right] = z.$$

■

# Quiz

Recall the four density forecasts (perfect, climatological, unfocused, sign-reversed). We simulate 10 000 forecast cases and plot a histogram of the PIT values for each forecast. For which forecasts do we expect a uniform PIT histogram?

A: only the perfect forecast

B: the perfect and the climatological forecasts

C: the perfect, the climatological and the unfocused forecasts

D: all four forecasts

# Example: Density forecasts (continued)

Recall that $Y \mid \mu \sim \mathcal{N}(\mu, 1)$ with $\mu \sim \mathcal{N}(0, 1)$, and $\mathbb{Q}(\tau = 1) = \mathbb{Q}(\tau = -1) = \frac{1}{2}$:

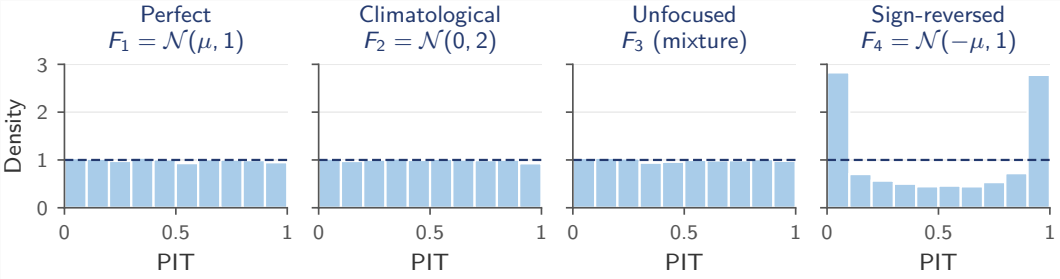
| Forecast | Predictive distribution | Marginally calibrated | Probabilistically calibrated | Ideal |
|---|---|---|---|---|
| Perfect | $F_1 = \mathcal{N}(\mu, 1)$ | Yes | Yes | Yes |
| Climatological | $F_2 = \mathcal{N}(0, 2)$ | Yes | Yes | Yes |
| Unfocused | $F_3 = \frac{1}{2}\left(\mathcal{N}(\mu, 1) + \mathcal{N}(\mu + \tau, 1)\right)$ | No | Yes | No |
| Sign-reversed | $F_4 = \mathcal{N}(-\mu, 1)$ | Yes | No | No |

The perfect and the climatological forecast are ideal, and thus calibrated by Theorem 2.7.

The unfocused forecast is probabilistically but not marginally calibrated, and the sign-reversed forecast is marginally but not probabilistically calibrated. Hence, there is no information set relative to which they are ideal (proofs: Exercises).

# Example: Density forecasts (continued)

PIT histograms for a simulated sample of size 10 000:




The PIT histogram cannot distinguish between the perfect, the climatological and the unfocused forecast.

→ We need additional criteria: sharpness (Section 2.3) and proper scoring rules (Section 3).

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# PIT histograms

Probabilistic calibration is considered the most critical requirement for probabilistic forecasts that take the form of predictive CDFs.

Checks for the uniformity of the PIT have formed a cornerstone of density forecast evaluation (Dawid, 1984; Diebold et al., 1998).

In practice, one observes a sample

$$\{(F_{1j}, \ldots, F_{kj}, y_j) : j = 1, \ldots, J\}$$

from the joint distribution of the probabilistic forecasts and the observation, and the uniformity of the PIT is assessed empirically.

The prevalent way of doing this is by plotting histograms of the PIT values for the various forecasting methods.

# Interpreting PIT histograms

For a probabilistically calibrated forecast, the PIT histogram is uniform. Typical deviations from uniformity are:

- U-shaped: underdispersed predictive distributions, with prediction intervals that are too narrow on average
- hump-shaped (inverse U-shaped): overdispersed predictive distributions
- skewed: predictive distributions that are biased in their location

Example: Let $Y = X + \epsilon$, where $X$ and $\epsilon$ are independent standard normal random variables, and consider the Gaussian forecast

$$F_\sigma = \mathcal{N}(X, \sigma^2).$$

Then $F_\sigma$ is underdispersed if $\sigma < 1$, neutrally dispersed if $\sigma = 1$, and overdispersed if $\sigma > 1$. If $\sigma = 1$, then $F_\sigma$ is ideal and thus calibrated.

# Quiz

The PIT histogram of a forecasting method is decreasing from left to right, i.e., small PIT values are much more frequent than large ones. What does this indicate?

A: The predictive distributions are too narrow.

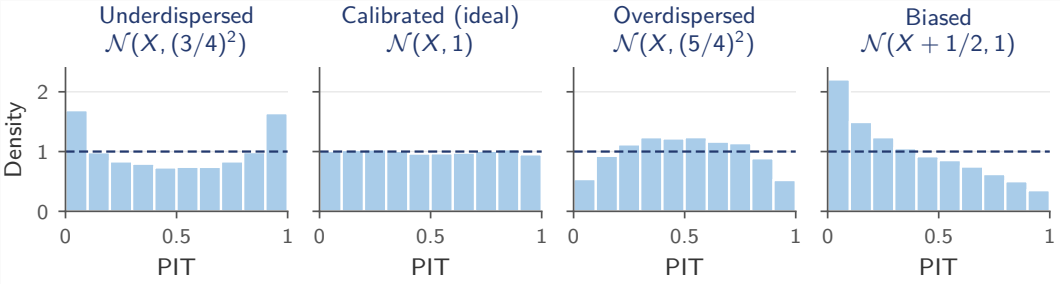
B: The predictive distributions are too wide.

C: The forecasts are systematically too high.

D: The forecasts are systematically too low.

# Example: PIT histograms

PIT histograms for a simulated sample of size 10 000 from the joint distribution of $Y$ and $F_\sigma$, and for a biased forecast $\mathcal{N}(X + \frac{1}{2}, 1)$:



# Rank histograms for ensemble forecasts

Frequently, probabilistic forecasts take the form of a discrete sample $x_1, \ldots, x_m$, where each sample value is considered an equally likely realization of $Y$.

A prominent example are ensemble weather forecasts (Section 1.2.6). The sample can be identified with its empirical CDF,

$$F(y) = \frac{1}{m} \sum_{i=1}^{m} \mathbf{1}\{x_i \leq y\}.$$

In principle, the (randomized) PIT can be computed as usual.

However, if the sample values and the observation come from continuous distributions, there is another, preferred way of checking calibration based on ranks.

# Rank histograms for ensemble forecasts

Let $x_{(1)} < \cdots < x_{(m)}$ denote the order statistics of $x_1, \ldots, x_m$, which divide the real line into $m+1$ bins. The rank of the observation $y$ is

$$r = \#\{i \in \{1, \ldots, m\} : x_i \leq y\} + 1 \in \{1, \ldots, m+1\}.$$




If the sample values and the observation are exchangeable, the observation falls into any of the $m+1$ bins with equal probability $\frac{1}{m+1}$.

Ranks are composited over forecast cases and their empirical frequencies are plotted in a rank histogram (or Talagrand diagram), which is interpreted in the same way as a PIT histogram.

# Quiz

An ensemble forecast has $m = 9$ members, and the ensemble members and the observation are exchangeable. With which probability does the observation fall outside the ensemble range $[x_{(1)}, x_{(m)}]$?

A: $\frac{1}{10}$

B: $\frac{1}{9}$

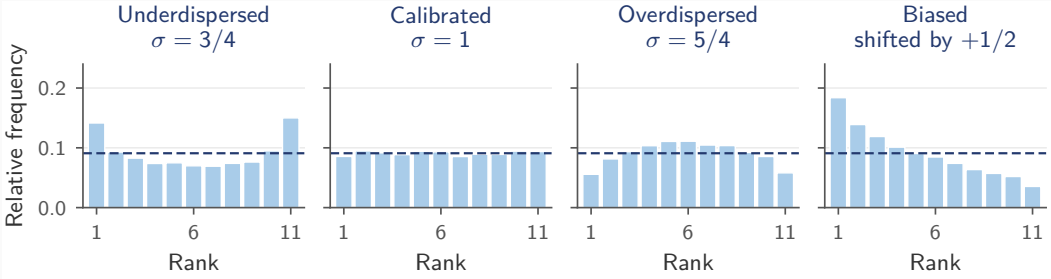
C: $\frac{2}{10}$

D: 0

# Example: Rank histograms

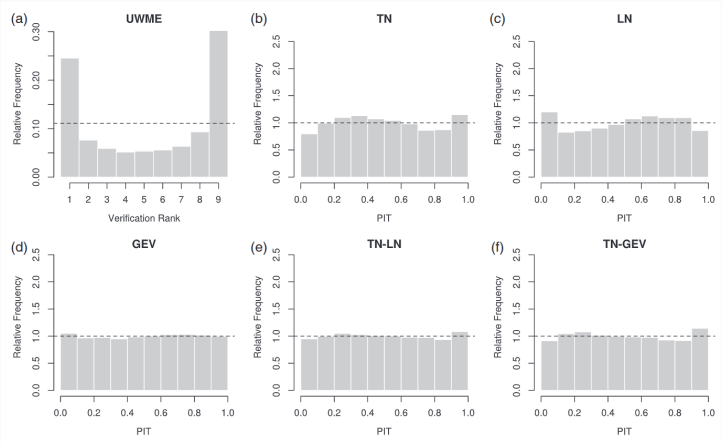
In particular, the ensemble range $[x_{(1)}, x_{(m)}]$ provides a central prediction interval with nominal coverage $\frac{m-1}{m+1}$.

Rank histograms for $m = 10$ members from $\mathcal{N}(X, \sigma^2)$ or $\mathcal{N}(X + \frac{1}{2}, 1)$, where $Y = X + \epsilon$ as before:



# Example: Rank histograms

Raw ensemble weather forecasts are often underdispersed and biased. Interpretation and pitfalls: Hamill (2001). This can be corrected by post-processing:




Baran and Lerch (2016)

# Calibration checks for time series forecasts

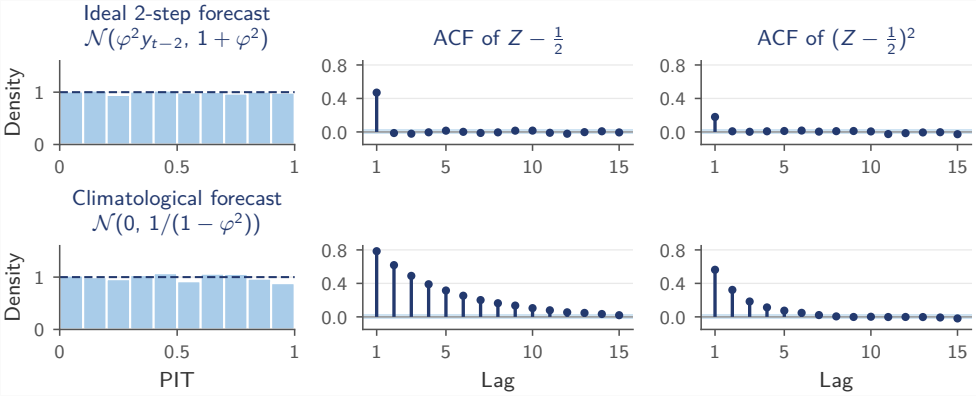
In time series settings, the observations are sequential, and the predictive distributions correspond to sequential $k$-step ahead forecasts.

The PIT values of ideal $k$-step ahead forecasts are independent at lags $\geq k$ (Diebold et al., 1998). Checks for uniformity can thus be supplemented by checks for independence, e.g., using sample autocorrelation functions (ACFs) of $Z - \frac{1}{2}$ and its powers.

Example: AR(1) process $Y_t = \varphi Y_{t-1} + \epsilon_t$ with $\varphi = 0.8$ and iid $\epsilon_t \sim \mathcal{N}(0, 1)$, and forecasts at lead time $k = 2$:

- ideal forecast given $y_{t-2}, y_{t-3}, \ldots$: $\mathcal{N}(\varphi^2 y_{t-2},\, 1 + \varphi^2)$
- climatological forecast: $\mathcal{N}\left(0,\, 1/(1 - \varphi^2)\right)$, ideal relative to the trivial $\sigma$-algebra

# Example: 2-step ahead forecasts of an AR(1) process

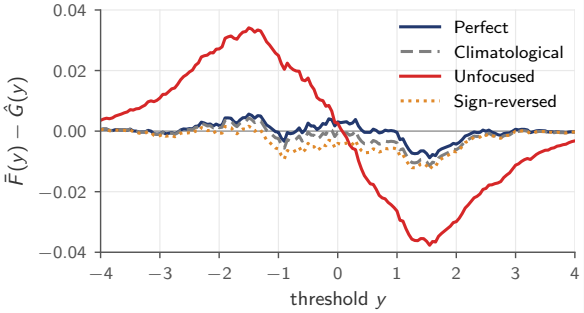



Both forecasts are probabilistically calibrated. The PIT values of the ideal forecast are correlated at lag 1 only, while those of the climatological forecast are strongly autocorrelated at lags $\geq 2$: it ignores available information.

# Checking marginal calibration

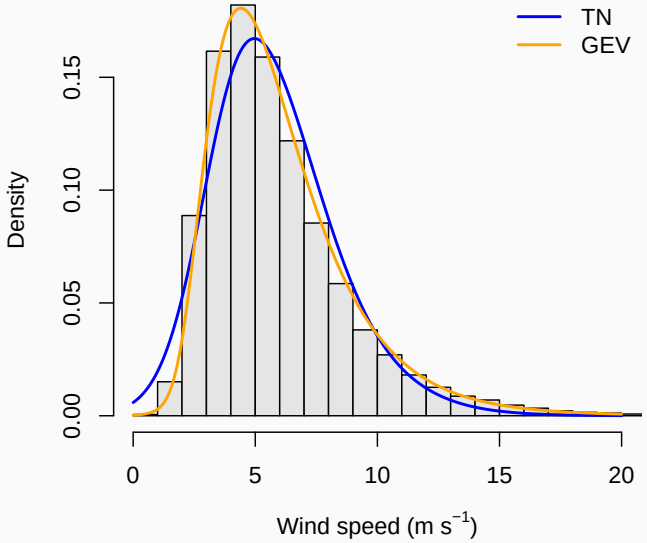
Marginal calibration concerns the equality of the forecast climate and the actual climate. To assess it, we can compare the average predictive CDF to the empirical CDF of the observations,

$$\bar{F}_J(y) = \frac{1}{J}\sum_{j=1}^{J} F_j(y) \qquad \text{vs.} \qquad \hat{G}_J(y) = \frac{1}{J}\sum_{j=1}^{J} \mathbf{1}\{y_j \le y\}.$$




Density forecasts example ($J = 10\,000$): only the unfocused forecast shows systematic deviations (cf. Gneiting et al., 2007).

# Example: Choice of parametric forecast distribution in post-processing




based on data and models from Lerch and Thorarinsdottir (2013)

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# Probability forecasts of a binary event

Recall that for a binary outcome $Y \in \{0, 1\}$, we identify the CDF-valued random quantity

$$F(y) = (1 - p)\mathbf{1}\{y \geq 0\} + p\mathbf{1}\{y \geq 1\}$$

with the probability forecast $p$ for a success, i.e., $Y = 1$.

**Definition 2.8: Conditional calibration**

The probability forecast $p$ is **conditionally calibrated** if

$$\mathbb{Q}(Y = 1 \mid p) = p \quad \text{almost surely.}$$

For example, among all cases in which a 30% chance of rain is forecast, it rains in 30% of the cases.

# Calibration for binary events

If the outcome is binary, the notions of probabilistic calibration and conditional calibration are equivalent.

**Theorem 2.9**

Consider a prediction space with a binary outcome $Y \in \{0, 1\}$, and a forecast $F$ that is identified with the probability forecast $p$. Then the following statements are equivalent:

(i) $F$ is probabilistically calibrated, i.e., $Z_F$ is uniformly distributed on the unit interval.

(ii) $p$ is conditionally calibrated, i.e., $\mathbb{Q}(Y = 1 \mid p) = p$ almost surely.

(iii) $F$ is ideal relative to the $\sigma$-algebra $\sigma(p)$ generated by $p$.

# Proof idea

(ii) $\Leftrightarrow$ (iii) holds by definition, and (iii) $\Rightarrow$ (i) follows from Theorem 2.7. It remains to show that (i) $\Rightarrow$ (ii).

Let $q(x) = \mathbb{Q}(Y = 1 \mid p = x)$. Given $p = x$, the randomized PIT is

$$Z_F = \begin{cases} V(1-x) \sim \mathcal{U}(0, 1-x) & \text{if } Y = 0 \quad (\text{probability } 1 - q(x)), \\ 1 - x + Vx \sim \mathcal{U}(1-x, 1) & \text{if } Y = 1 \quad (\text{probability } q(x)). \end{cases}$$

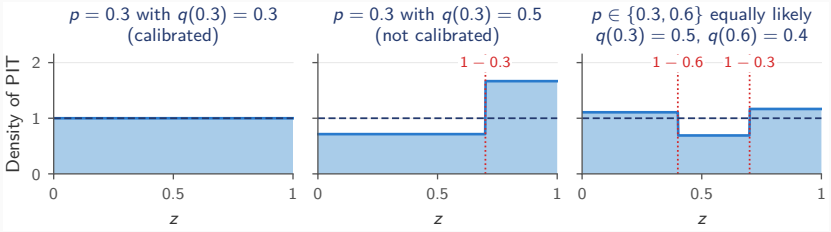
Hence, the conditional density of $Z_F$ given $p = x$ is a step function with a jump at $1 - x$,

$$u(z \mid x) = \frac{1 - q(x)}{1 - x} \quad \text{for } z < 1 - x, \qquad u(z \mid x) = \frac{q(x)}{x} \quad \text{for } z > 1 - x,$$

and it is flat (i.e., uniform) if and only if $q(x) = x$.

# Proof idea

Key idea: Suppose $p$ takes finitely many values $x_1, \ldots, x_n \in (0,1)$ with probabilities $h_1, \ldots, h_n$. Then the density $u(z) = \sum_j h_j\, u(z \mid x_j)$ of $Z_F$ is piecewise constant, with a jump of size $h_j\,(q(x_j) - x_j)/(x_j(1-x_j))$ at $z = 1 - x_j$.




If $Z_F$ is uniform, there are no jumps, so $q(x_j) = x_j$ for all $j$, which is (ii). ■

General case: the same argument applies to arbitrary intervals of forecast values. Point masses of $p$ at 0 or 1 are treated separately.

# Reliability diagrams

Theorem 2.9 draws a connection from the PIT histogram to the reliability diagram or calibration curve, which is the key diagnostic tool for assessing the calibration of probability forecasts for a binary event (Murphy and Winkler, 1992; Ranjan and Gneiting, 2010).

A reliability diagram plots conditional event frequencies against binned forecast probabilities, with deviations from the diagonal indicating violations of conditional calibration.

Given a sample $\{(p_j, y_j) : j = 1, \ldots, J\}$ and a partition $0 = b_0 < b_1 < \cdots < b_I = 1$ of the unit interval into bins, we plot

$$\frac{\#\{j : p_j \in [b_{i-1}, b_i),\, y_j = 1\}}{\#\{j : p_j \in [b_{i-1}, b_i)\}} \qquad \text{vs.} \qquad \frac{1}{2}(b_{i-1} + b_i)$$

for $i = 1, \ldots, I$, and compare the resulting graph to the diagonal.

# Example: Binary events

Consider a binary event $Y$ with success probability $\Phi(a_1 + a_2)$, i.e.,

$$Y \mid a_1, a_2 \sim \text{Ber}\left(\Phi(a_1 + a_2)\right),$$

where $a_1 \sim \mathcal{N}(0, \sigma_1^2)$ and $a_2 \sim \mathcal{N}(0, \sigma_2^2)$ are independent, and $\Phi$ denotes the standard normal CDF (Ranjan and Gneiting, 2010).

Three forecasters have access to different information sets:

- forecaster 1 knows $a_1$ and issues $p_1 = \Phi\big(a_1/\sqrt{1+\sigma_2^2}\big)$,
- forecaster 2 knows $a_2$ and issues $p_2 = \Phi\big(a_2/\sqrt{1+\sigma_1^2}\big)$,
- forecaster 3 knows $a_1$ and $a_2$, and issues $p_3 = \Phi(a_1 + a_2)$.

Each forecast is ideal relative to the respective information set, and thus calibrated (Exercise).

Hint: If $X \sim \mathcal{N}(\mu, \tau^2)$, then $\mathbb{E}\,\Phi(X) = \Phi\big(\mu/\sqrt{1+\tau^2}\big)$.

# Quiz

A reliability diagram lies below the diagonal for high forecast probabilities, and above the diagonal for low forecast probabilities. The probability forecasts are

A: calibrated

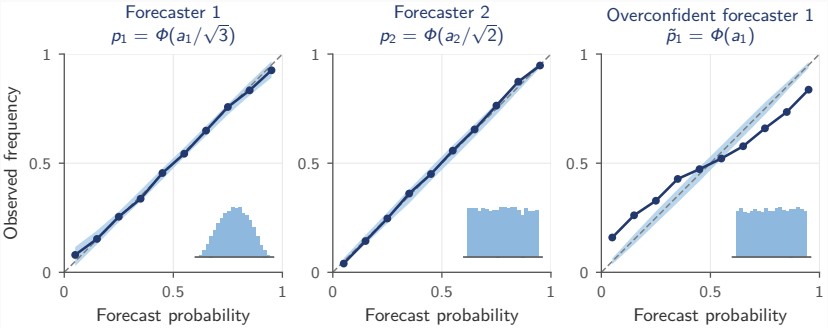
B: overconfident, i.e., too extreme

C: underconfident, i.e., not extreme enough

D: biased towards the event $Y = 1$

# Example: Reliability diagrams

Reliability diagrams for a sample of size 10 000 with $\sigma_1^2 = 1$ and $\sigma_2^2 = 2$, and for an overconfident version $\tilde{p}_1 = \Phi(a_1)$ of forecaster 1, who ignores the uncertainty about $a_2$:




Shaded: 95% consistency bands under the hypothesis of calibration, obtained by resampling (Bröcker and Smith, 2007). Insets: histograms of the forecast values.

# CORP reliability diagrams

Binning and counting can be unstable: results depend on the (ad hoc) choice of the bins, particularly for small samples or discrete forecasts.

The CORP approach (Dimitriadis et al., 2021) avoids ad hoc choices and is

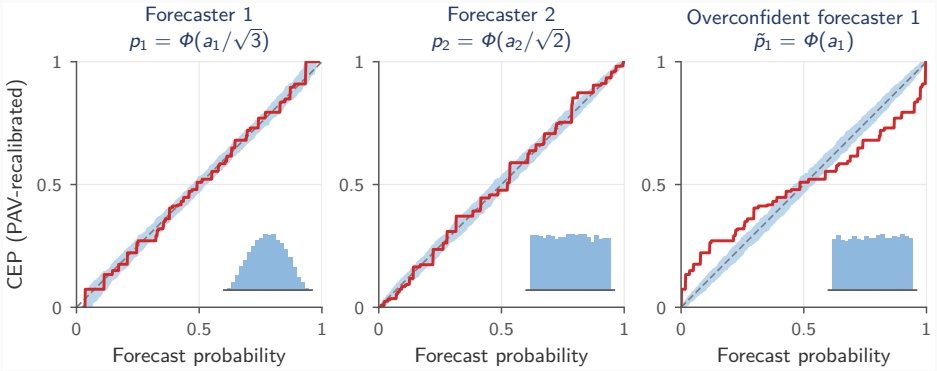
- Consistent,
- Optimally binned,
- Reproducible, and
- based on the Pool-adjacent-violators (PAV) algorithm.

The conditional event probabilities (CEPs) are estimated via isotonic regression of the binary outcomes on the forecast values, which is computed with the PAV algorithm.

Implementation: `R` package `reliabilitydiag`

# Example: CORP reliability diagrams

CORP reliability diagrams for the three forecasts from before:




Shaded: 90% consistency bands under the hypothesis of calibration, obtained by resampling. Insets: histograms of the forecast values.

# Auto-calibration

For non-binary outcomes, the natural generalization of conditional calibration is auto-calibration (Tsyplakov, 2011).

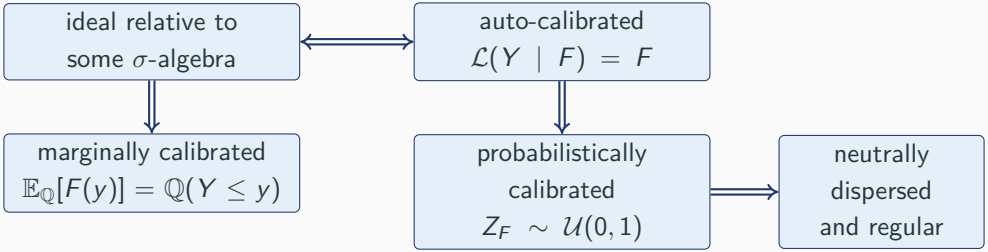
**Definition 2.10: Auto-calibration**

The forecast $F$ is **auto-calibrated** if

$$\mathcal{L}(Y \mid F) = F \quad \text{almost surely.}$$

- A forecast is auto-calibrated if and only if it is ideal relative to some $\sigma$-algebra, namely, $\sigma(F)$ (Exercise).
- For a binary outcome, auto-calibration is equivalent to conditional calibration.
- For non-binary outcomes, the equivalence to probabilistic calibration fails, and auto-calibration is a much stronger requirement than probabilistic calibration.

# Summary: Notions of calibration




- The converse implications fail in general, as shown by the unfocused and sign-reversed forecasts.
- For a binary outcome, probabilistic, conditional and auto-calibration are equivalent.
- The climatological forecast is calibrated in all of these senses, but not very informative $\rightarrow$ sharpness (Section 2.3)

# Overview

2. Prediction Spaces, Calibration and Sharpness

2.1 Prediction spaces

2.2 Calibration

2.3 Sharpness

# Calibration and sharpness

Following Murphy and Winkler (1987) and Gneiting et al. (2007), the goal in probabilistic forecasting is to

maximize the sharpness of the predictive distributions subject to calibration.

- Calibration is a multi-faceted property of the joint distribution of the forecasts and the observations, with auto-calibration being the strongest such property. Essentially, the observation is supposed to be a random draw from the predictive distribution.
- Sharpness refers to the concentration of the predictive distributions, and so is a property of the forecasts only.

  Evidently, the more concentrated, the better, subject to calibration.

# Quiz

Which of the following properties of a forecasting method can be assessed without observing the outcomes?

A: marginal calibration

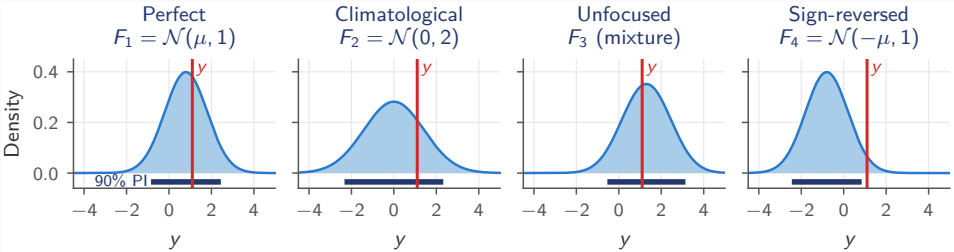
B: probabilistic calibration

C: dispersion, i.e., $\text{Var}(Z_F)$

D: sharpness

# Example: Density forecasts (continued)

Predictive densities for a single forecast case ($\mu = 0.8$, $\tau = 1$), with the observation $y$ (red) and the central 90% prediction interval (PI, dark blue):




The perfect, climatological and unfocused forecasts are all probabilistically calibrated, but the perfect forecast is the sharpest.

The sign-reversed forecast is equally sharp, but not calibrated.

# Sharpness of probability forecasts

In the case of probability forecasts of a binary event, a sharp forecast tends to be close to 0 or 1.

Recall the binary example from Section 2.2.4 with $\sigma_1^2 = 1$ and $\sigma_2^2 = 2$, where

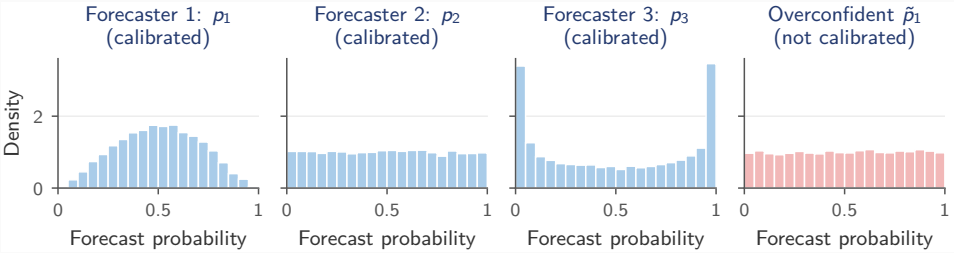
- forecaster 1 knows $a_1$ and issues $p_1 = \Phi\big(a_1/\sqrt{1+\sigma_2^2}\big)$,
- forecaster 2 knows $a_2$ and issues $p_2 = \Phi\big(a_2/\sqrt{1+\sigma_1^2}\big)$,
- forecaster 3 knows $a_1$ and $a_2$, and issues $p_3 = \Phi(a_1 + a_2)$,

and $\tilde{p}_1 = \Phi(a_1)$ is an overconfident version of forecaster 1.

Forecasts $p_1$, $p_2$ and $p_3$ are ideal, and thus calibrated.

# Example: Sharpness of probability forecasts

Histograms of the forecast probabilities for a sample of size 10 000:




All three ideal forecasts are calibrated, but $p_3$ is sharper than $p_2$, which is sharper than $p_1$: the more information, the sharper the ideal forecast.

The overconfident forecast $\tilde{p}_1$ is as sharp as $p_2$, but not calibrated (cf. its reliability diagram in Section 2.2.4).

# Sharpness of density forecasts

In the case of density forecasts of a real-valued variable, we can assess sharpness in terms of the width of prediction intervals.

For example, the central 90% prediction interval ought to be as short as possible, subject to its empirical coverage being at the nominal 90% level, with 5% of the observations lying to its left and right, respectively.

Competing forecasts can be compared by the average width of their prediction intervals or, if the widths vary between forecast cases, by boxplots of the widths (sharpness diagram).

For a continuous predictive CDF $F$, the observation falls into the central $(1 - \alpha)$ prediction interval if and only if $Z_F \in [\frac{\alpha}{2}, 1 - \frac{\alpha}{2}]$. Hence, probabilistically calibrated forecasts attain the nominal coverage.

# Example: Density forecasts (continued)

Width and coverage of the central 50% and 90% prediction intervals:

| | 50% interval | | 90% interval | |
|---|---|---|---|---|
| Forecast | Width | Coverage | Width | Coverage |
| Perfect | 1.35 | 50.0% | 3.29 | 90.0% |
| Climatological | 1.91 | 50.0% | 4.65 | 90.0% |
| Unfocused | 1.52 | 50.0% | 3.68 | 90.0% |
| Sign-reversed | 1.35 | 23.7% | 3.29 | 53.8% |

Among the calibrated forecasts, the perfect forecast is the sharpest. The sign-reversed forecast is equally sharp, but its prediction intervals are far from the nominal coverage.

# Example: 2-step ahead forecasts of an AR(1) process (continued)

We return to the AR(1) process $Y_t = \varphi Y_{t-1} + \epsilon_t$ with $\varphi = 0.8$ and iid $\epsilon_t \sim \mathcal{N}(0, 1)$ from Section 2.2.3, and compare four forecasts at lead time $k = 2$:

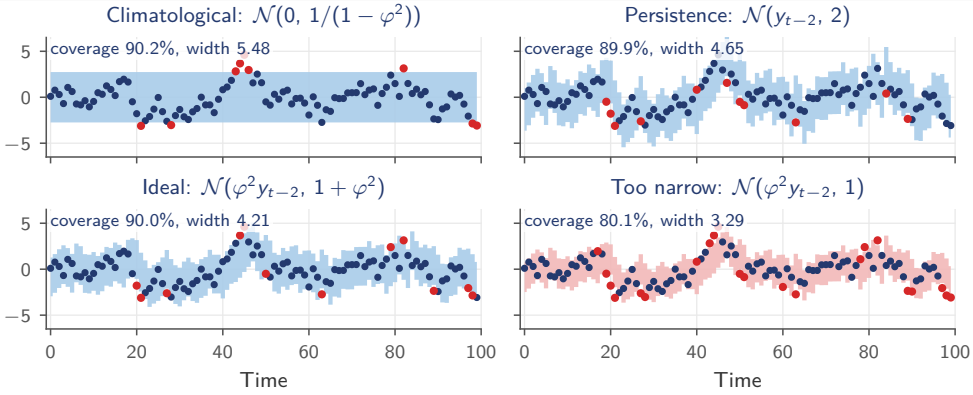
- climatological: $\mathcal{N}\left(0, 1/(1-\varphi^2)\right)$
- persistence (probabilistic no-change): $\mathcal{N}(y_{t-2}, 2)$
- ideal: $\mathcal{N}(\varphi^2 y_{t-2}, 1+\varphi^2)$
- too narrow: $\mathcal{N}(\varphi^2 y_{t-2}, 1)$, which ignores the uncertainty due to $\epsilon_{t-1}$

The climatological, persistence and ideal forecasts are probabilistically calibrated.

For the persistence forecast, use that $Y_t - Y_{t-2} \sim \mathcal{N}(0, 2)$ (Exercise).

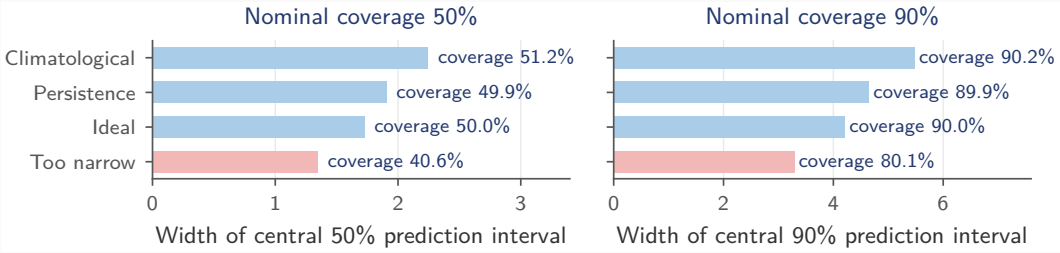
# Example: 2-step ahead forecasts of an AR(1) process (continued)

Central 90% prediction intervals for 100 time steps (observations outside the interval in red); coverage and width are computed from 5 000 forecast cases:



# Example: 2-step ahead forecasts of an AR(1) process (continued)

Width and empirical coverage of the central 50% and 90% prediction intervals:




The climatological, persistence and ideal forecasts are (empirically) calibrated, and the ideal forecast is the sharpest.

The "too narrow" forecast is even sharper, but its prediction intervals fall short of the nominal coverage.

# Quiz

Forecast A issues central 90% prediction intervals with an average width of 3.3 and an empirical coverage of 80%. For forecast B, the average width is 4.2 and the empirical coverage is 90%. Which statement is correct?

A: Forecast A is preferable, since it is sharper.

B: Forecast B is preferable, since forecast A is not calibrated.

C: Both forecasts are equally good.

D: Neither, since the climatological forecast is always preferable.

# Calibration and sharpness: Summary

| | |
|---|---|
| **calibrated and sharp**<br>perfect forecast, ideal AR forecast, forecaster 3 in the binary example | **calibrated, but not sharp**<br>climatological forecasts |
| **sharp, but not calibrated**<br>sign-reversed forecast, overconfident forecasts | **neither**<br>e.g., overdispersed forecasts |

To address calibration and sharpness simultaneously, we use proper scoring rules (Gneiting and Raftery, 2007), to be discussed in Section 3.